

SMART METRO TICKET GENERATION SYSTEM

Phase 1.3 — Empathize: Empathy Map Canvas

Date	
Team ID	
Project Name	Smart Metro Ticket Generation System
Phase	Phase 1.3 — Empathize: Empathy Map Canvas
Maximum Marks	4 Marks

PHASE 1.3

1. Introduction to Empathy Mapping

Empathy mapping is the first step of the design-thinking process. Before a solution can be designed, the team must deeply understand the people who will use it. For the Smart Metro Ticket Generation System, this means stepping into the shoes of a daily metro commuter — observing their environment, listening to their conversations, and understanding what they think, feel, say, and do while trying to book and use a metro ticket. The Empathy Map Canvas is a simple, visual tool that organizes these observations into six segments, helping the team build genuine understanding rather than relying on assumptions.

2. Purpose & Methodology

The empathy map was built by combining direct observation at metro stations, short conversations with regular commuters, and analysis of common complaints shared on public forums and station feedback registers. Observations focused on peak-hour ticket counters, smart-card recharge kiosks, and station entry/exit gates, since these are the touch-points where friction is most visible.

- On-ground observation of commuter behaviour at ticket counters and gates.
- Short, informal interviews with commuters of different age groups and travel frequency.
- Review of commonly reported complaints about queues, recharge, and payment failures.
- Grouping of findings into the six empathy map segments for analysis.

3. Persona Overview

Name Represents a typical daily metro rider (composite persona).	Role Working professional / student who commutes on the metro daily.	Goal Reach the platform and board the train quickly, without delays.	Frustration Long queues and confusing recharge steps during peak hours.
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4. Understanding the Six Segments

Each quadrant of the empathy map captures a different dimension of the commuter's experience. Together, they form a complete picture of the user's world, which guides the team toward a problem statement that is grounded in real behaviour rather than guesswork.

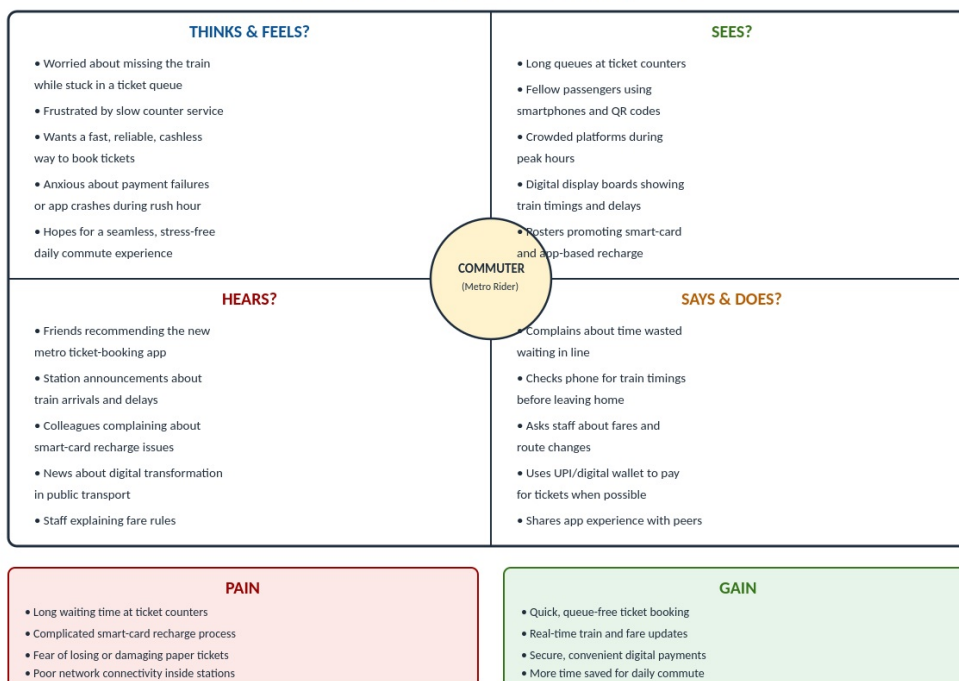
Segment	What It Captures
Thinks & Feels	Internal worries, hopes, and emotions the commuter rarely says out loud — such as anxiety about missing a train.
Sees	What the commuter observes in their environment — queues, crowded platforms, digital display boards.
Hears	What friends, announcements, staff, and media tell the commuter about the metro and ticketing options.
Says & Does	Observable actions and statements — checking timings, asking staff, paying via UPI.
Pain	Fears, frustrations, and obstacles that stand between the commuter and a smooth journey.
Gain	Wants, needs, and measures of success the commuter hopes to achieve.

5. Empathy Map Canvas

The completed canvas below consolidates all research findings for the daily metro commuter persona into the six segments described above.

EMPATHY MAP CANVAS

Smart Metro Ticket Booking System — User: Daily Metro Commuter



6. Pain Points & Gains Summary

Reviewing the empathy map as a whole surfaces a clear set of pains to solve for and gains to design toward. These form the direct input to the Problem Statement developed in the next phase.

PAIN

- Long waiting time at ticket counters during peak hours
- Complicated, multi-step smart-card recharge process
- Fear of losing or damaging paper tickets
- Poor network connectivity inside underground stations

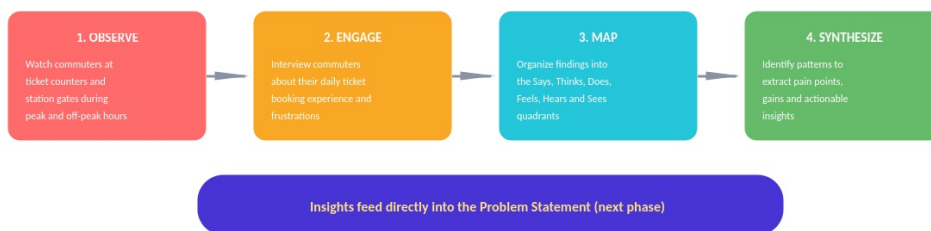
GAIN

- Quick, queue-free ticket booking experience
- Real-time train and fare updates on the go
- Secure, convenient digital payment options
- More time saved for the daily commute

7. From Empathy to Insight

The diagram below shows how raw observations and conversations are systematically converted into actionable insight, ensuring the empathy work directly informs the next design-thinking phase.

FROM EMPATHY TO INSIGHT — PROCESS FLOW



8. Key Takeaways

- Commuters value speed and predictability above all else during their daily commute.
- Emotional friction (anxiety, frustration) is often triggered by process complexity, not just wait time.
- Digital literacy varies widely, so any solution must remain simple for first-time app users.
- Real-time information reduces uncertainty and improves the overall commuting experience.